Python Programming Practice Programs

1. Check Even or Odd

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print(f"{num} is an even number.")
else:
    print(f"{num} is an odd number.")
```

2. Positive, Negative or Zero

```
num = float(input("Enter a number: "))

if num > 0:
    print(f"{num} is a positive number.")
elif num < 0:
    print(f"{num} is a negative number.")
else:
    print("The number is zero.")
```

3. Largest of Two Numbers

```
num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))

if num1 > num2:
    print(f"The largest number is {num1}.")
elif num2 > num1:
    print(f"The largest number is {num2}.")
else:
    print("Both numbers are equal.")
```

4. Largest of Three Numbers

```
num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))
num3 = float(input("Enter the third number: "))

if num1 >= num2 and num1 >= num3:
    largest = num1
elif num2 >= num1 and num2 >= num3:
    largest = num2
```

```python
else:
    largest = num3

print(f"The largest number is {largest}.")
```

5. Leap Year Check

```python
year = int(input("Enter a year: "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print(f"{year} is a leap year.")
else:
    print(f"{year} is not a leap year.")
```

6. Print Numbers 1 to N

```python
n = int(input("Enter a number: "))

print(f"Numbers from 1 to {n}:")
for i in range(1, n + 1):
    print(i)
```

7. Multiplication Table

```python
num = int(input("Enter a number: "))

print(f"Multiplication table for {num}:")
for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")
```

8. Sum of First N Natural Numbers

```python
n = int(input("Enter a number: "))

if n < 1:
    print("Please enter a positive number.")
else:
    total = 0
    for i in range(1, n + 1):
        total += i
    print(f"The sum of the first {n} natural numbers is {total}.")
```

9. Factorial of a Number

```python
num = int(input("Enter a number: "))
```

```python
if num < 0:
    print("Factorial is not defined for negative numbers.")
else:
    factorial = 1
    for i in range(1, num + 1):
        factorial *= i
    print(f"The factorial of {num} is {factorial}.")
```

10. Print Even Numbers 1 to 50

```python
print("Even numbers between 1 and 50:")
for i in range(1, 51):
    if i % 2 == 0:
        print(i)
```

11. Prime Number Check

```python
num = int(input("Enter a number: "))

if num <= 1:
    print(f"{num} is not a prime number.")
else:
    is_prime = True
    for i in range(2, int(num**0.5) + 1):
        if num % i == 0:
            is_prime = False
            break
    if is_prime:
        print(f"{num} is a prime number.")
    else:
        print(f"{num} is not a prime number.")
```

12. Armstrong Number Check

```python
num = int(input("Enter a number: "))
num_str = str(num)
num_digits = len(num_str)

sum_of_powers = 0
temp = num

while temp > 0:
    digit = temp % 10
    sum_of_powers += digit ** num_digits
```

```python
    temp //= 10

if sum_of_powers == num:
    print(f"{num} is an Armstrong number.")
else:
    print(f"{num} is not an Armstrong number.")
```

13. Sum of Digits

```python
num = int(input("Enter a number: "))
temp = abs(num)
digit_sum = 0

while temp > 0:
    digit_sum += temp % 10
    temp //= 10

print(f"The sum of digits of {num} is {digit_sum}.")
```

14. Reverse a Number

```python
num = int(input("Enter a number: "))
reverse = 0
temp = abs(num)

while temp > 0:
    digit = temp % 10
    reverse = reverse * 10 + digit
    temp //= 10

if num < 0:
    reverse = -reverse

print(f"The reverse of {num} is {reverse}.")
```

15. Palindrome Number Check

```python
num = int(input("Enter a number: "))
original = str(num)
reversed_num = original[::-1]

if original == reversed_num:
    print(f"{num} is a palindrome number.")
else:
    print(f"{num} is not a palindrome number.")
```

16. Star Pattern N Rows

```
n = int(input("Enter the number of rows: "))

for i in range(1, n + 1):
    print("* " * i)
```

17. Number Pattern N Rows

```
n = int(input("Enter the number of rows: "))

for i in range(1, n + 1):
    for j in range(1, i + 1):
        print(j, end=" ")
    print()
```

18. Largest Number in List

```
numbers = input("Enter numbers separated by spaces: ").split()
numbers = [float(num) for num in numbers]

largest = max(numbers)
print(f"The largest number in the list is {largest}.")
```

19. Sum of Elements in List

```
numbers = input("Enter numbers separated by spaces: ").split()
numbers = [float(num) for num in numbers]

total = sum(numbers)
print(f"The sum of all elements in the list is {total}.")
```

20. Palindrome String Check

```
text = input("Enter a string: ")
cleaned = text.replace(" ", "").lower()
reversed_text = cleaned[::-1]

if cleaned == reversed_text:
    print(f'"{text}" is a palindrome.')
else:
    print(f'"{text}" is not a palindrome.')
```

21. Count Vowels and Consonants

```python
text = input("Enter a string: ")
text = text.lower()

vowels = "aeiou"
vowel_count = 0
consonant_count = 0

for char in text:
    if char.isalpha():
        if char in vowels:
            vowel_count += 1
        else:
            consonant_count += 1

print(f"Vowels: {vowel_count}")
print(f"Consonants: {consonant_count}")
```

22. Factorial Function

```python
def factorial(n):
    if n < 0:
        return None
    result = 1
    for i in range(1, n + 1):
        result *= i
    return result

num = int(input("Enter a number: "))

fact = factorial(num)
if fact is None:
    print("Factorial is not defined for negative numbers.")
else:
    print(f"The factorial of {num} is {fact}.")
```

23. Prime Check Function

```python
def is_prime(n):
    if n <= 1:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True
```

```
num = int(input("Enter a number: "))

if is_prime(num):
    print(f"{num} is a prime number.")
else:
    print(f"{num} is not a prime number.")
```

24. Fibonacci Series

```
n = int(input("Enter the number of terms: "))

if n <= 0:
    print("Please enter a positive integer.")
elif n == 1:
    print("Fibonacci series up to 1 term:")
    print(0)
else:
    print(f"Fibonacci series up to {n} terms:")
    a, b = 0, 1
    print(a, end=" ")
    for _ in range(1, n):
        print(b, end=" ")
        a, b = b, a + b
    print()
```

25. Strong Number Check

```
def factorial(n):
    result = 1
    for i in range(1, n + 1):
        result *= i
    return result

num = int(input("Enter a number: "))
temp = num
sum_factorials = 0

while temp > 0:
    digit = temp % 10
    sum_factorials += factorial(digit)
    temp //= 10

if sum_factorials == num:
    print(f"{num} is a Strong number.")
else:
```

```
    print(f"{num} is not a Strong number.")
```

26. Perfect Number Check

```
num = int(input("Enter a number: "))

if num <= 1:
    print(f"{num} is not a Perfect number.")
else:
    divisor_sum = 0
    for i in range(1, num):
        if num % i == 0:
            divisor_sum += i

    if divisor_sum == num:
        print(f"{num} is a Perfect number.")
    else:
        print(f"{num} is not a Perfect number.")
```

27. Write and Read Message From File

```
message = input("Enter a message: ")

with open("message.txt", "w") as file:
    file.write(message)

with open("message.txt", "r") as file:
    read_message = file.read()

print("Message read from the file:")
print(read_message)
```